Schrodinger - Hamiltonian Mechanics

$p = \frac{h}{\lambda} \qquad k = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{2\pi}{k}$; λ [m], k [rad/m] (spatial frequency)

$p = \frac{h}{\left(\frac{2\pi}{k}\right)} = \frac{hk}{2\pi}$; $\hbar = \frac{h}{2\pi}$

$\Psi = \hat{\Psi}(\vec{k}) e^{i\vec{k}\cdot\vec{r}}$

$p = \hbar k$

$\vec{p} = \hbar \vec{k}$

$ik_x = \frac{\partial}{\partial x}$

$ik_y = \frac{\partial}{\partial y}$

$i \begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix}$

$\therefore \hat{p} = \hbar(-i\nabla)$

$\hat{p} = -i\hbar\nabla$

$i\vec{k} = \nabla$

$-i(i\vec{k}) = -i\nabla$

$\vec{k} = -i\nabla$

$T = \frac{\vec{p}\cdot\vec{p}}{2m}$, T is translational kinetic energy

$\hat{T}_e = \frac{\hat{p}\cdot\hat{p}}{2m_e}$ for electron

$T = \frac{1}{2}mv^2$, In newtonian mechanics, $\vec{p} = m\vec{v}$

$\hat{T}_e = \frac{(-i\hbar\nabla)\cdot(-i\hbar\nabla)}{2m_e} = \frac{i^2\hbar^2\nabla^2}{2m_e} = -\frac{\hbar^2\nabla^2}{2m_e}$

$$\boxed{\hat{T}_e = -\frac{\hbar^2\nabla^2}{2m_e}}$$

Schrodinger-Hamiltonian Mechanics

electron 1 is characterized by Ψ_1

electron 2 is characterized by Ψ_2

⋮

electron j is characterized by Ψ_j

⋮

electron N is characterized by Ψ_N

There are N number of electrons.

U_1 is potential energy stored in electron 1

⋮

U_j is potential energy stored in electron j

⋮

U_N is potential energy stored in electron N

Schrodinger and Hamiltonian Mechanics

$$U_j = U_{j,\text{attraction}} + U_{j,\text{repulsion}}$$

$$U_{j,\text{attraction}} = \frac{(Ze)(-e)}{4\pi\varepsilon_0 r_j}$$

e is charge of proton

-e is charge of electron

Z is atomic number (number of protons)

$$U_{j,\text{attraction}} = -\frac{Ze^2}{4\pi\varepsilon_0 r_j}$$

$$U_{j,\text{repulsion}} = \sum_{\substack{k=1 \\ k\neq j}}^{N} \frac{(-e)(-e)}{4\pi\varepsilon_0 |\vec{r}_j - \vec{r}_k|}$$

$$U_{j,\text{repulsion}} = \sum_{\substack{k=1 \\ k\neq j}}^{N} \frac{e^2}{4\pi\varepsilon_0 |\vec{r}_j - \vec{r}_k|}$$

$$U_j = -\frac{Ze^2}{4\pi\varepsilon_0 r_j} + \sum_{\substack{k=1 \\ k\neq j}}^{N} \frac{e^2}{4\pi\varepsilon_0 |\vec{r}_j - \vec{r}_k|}$$

Schrodinger - Hamiltonian Mechanics

$$E = T + U$$

$$E = hf \; ; \; \underbrace{f\,[1/s]}_{\text{temporal frequency}} \qquad \omega\,[rad/s]$$

$$\omega = 2\pi f$$

$$f = \frac{\omega}{2\pi}$$

$$E = \frac{h\omega}{2\pi} \; ; \; \hbar = \frac{h}{2\pi}$$

$$\hat{E} = \hbar\left(i\frac{\partial}{\partial t}\right)$$

$$\boxed{\hat{E} = i\hbar\frac{\partial}{\partial t}}$$

$$E = \hbar\omega$$

$$\psi = \hat{\psi}(\vec{k}, t)\, e^{i(\vec{k}\cdot\vec{r} - \omega t)}$$

$$\frac{\partial \psi}{\partial t} = -i\omega\,\hat{\psi}(\vec{k}, t)\, e^{i(\vec{k}\cdot\vec{r} - \omega t)}$$

$$\frac{\partial \psi}{\partial t} = -i\omega\psi$$

$$\frac{\partial}{\partial t} = -i\omega$$

$$i\frac{\partial}{\partial t} = -i^2\omega \rightarrow i\frac{\partial}{\partial t} = \hat{\omega}$$

Schrodinger - Hamiltonian Mechanics

So $\hat{T} = -\frac{\hbar^2 \nabla^2}{2m_e}$

and $\hat{E} = i\hbar \frac{\partial}{\partial t}$

$\hat{E} = \hat{T} + U_{electron}$

Multiply by $\Psi_{electron}$

$\hat{E}\Psi_{electron} = \hat{T}\Psi_{electron} + U_{electron}\Psi_{electron}$

Based on electron j from 1 to N where N is number of electrons

$\hat{E}\Psi_j = \hat{T}\Psi_j + U_j\Psi_j$

$\therefore i\hbar \frac{\partial \Psi_j}{\partial t} = -\frac{\hbar^2 \nabla^2 \Psi_j}{2m_e} + U_j \Psi_j$

And $U_j = -\frac{Ze^2}{4\pi\varepsilon_0 r_j} + \sum_{\substack{k=1 \\ k\neq j}}^{N} \frac{e^2}{4\pi\varepsilon_0 |\vec{r}_j - \vec{r}_k|}$

$\therefore i\hbar \frac{\partial \Psi_j}{\partial t} = -\frac{\hbar^2 \nabla^2 \Psi_j}{2m_e} + \Psi_j \left(-\frac{Ze^2}{4\pi\varepsilon_0 r_j} + \sum_{\substack{k=1 \\ k\neq j}} \frac{e^2}{4\pi\varepsilon_0 |\vec{r}_j - \vec{r}_k|} \right)$